

USEPA
DMR-QA 37

**NPDES PERMITTEE
DATA REPORT FORM**



ERA

A Waters Company

Due September 1, 2017

USEPA NPDES MI0020257

Permit #:

Permit Ext:

Permittee Name: Cadillac WWTP

Facility Address: 1121 Plett Rd.

City: Cadillac

State: MI

Postal Code: 49601

Phone Number: (231) 775-2368

Fax Number: (231) 775-2876

E-mail address: lab@cadillac-mi.net

Optional: If WP study was used, list PT provider name:

Optional: WP study number(s):

For DMRQA-37, conducted in 2017, the Permittee ensured that their laboratory(s) performing the required analyses:

Received PT Samples

Submitted Complete and Accurate Data
by July 7, 2017

Received a Graded Report by
August 4, 2017

Yes ☒ No ☐

Yes ☒ No ☐

Yes ☒ No ☐

Certification by Permit Holder or Authorized Representative

(as per 40 C.F.R. Section 122.22)

I certify under penalty of law that this document and all attachments were prepared under my direction or supervision in accordance with a system designed to assure that qualified personnel properly gather and evaluate the information submitted. Based on my inquiry of the person or persons who manage the system, or those persons directly responsible for gathering the information, the information submitted is, to the best of my knowledge and belief, true, accurate and complete. Each reported value was produced from a single analytical run using the analytical system that routinely performs these analyses to produce compliance monitoring data required under our National Pollutant Discharge Elimination System (NPDES) Permit. Neither I nor any of my subordinates compared our results from independent analyses conducted by us or any other laboratory before we reported our results to the USEPA. I am aware that there are significant penalties for submitting false information, including the possibility of fine and imprisonment for knowing violations.

Name of
Certifying Official: Cindy A Tomaszewski

Title: Lab Supervisor

Signature:

Date Signed:

08/13/2017

Mailing Address: 200 N Lake Street

(enter only if different
from address above)

City: Cadillac

State: MI

Postal Code: 49601

Phone Number: (231) 775-2368

E-mail address: lab@cadillac-mi.net

United States
ENVIRONMENTAL PROTECTION AGENCY
Laboratory Performance Evaluation
Laboratory DMR-QA Evaluation Study 37

USEPA NPDES Permit #: MI0020257

Permit Ext:

Identification of all CHEM, MICRO and TOX laboratories who did analyses for this permit							
Name of Laboratory	Address of Laboratory	U.S. EPA Lab Code	Lab Analysis Check box(es) that apply			Lab Type*	State-certified Lab**
			CHEM	MICRO	TOX		
Cadillac WWTP	200 N. Lake St. Cadillac, MI 49601	MI00082	☒	☒	☐	G	☐
Kar Labs	4425 Manchester Rd. Kalamazoo, MI 49001	MI00041	☒	☐	☐	C	☐
ENVIRONMENTAL RESOURCES MANAGEMENT	3352 128th Ave Holland MI 49424	MI01183			☒	C	

* Lab Types: C = Commercial F = Federal G = Local Government I = Industrial O = Other S = State

** See Footnote 2 on DMRQA-37, Frequently Asked Questions page

Permittee Name: Cadillac WWTP

Permit Number: MI0020257

EPA Lab Code: MI00041

Chemistry/Microbiology Analyte Checklist

DMRQA Study 37

Analyte Test / Method	Test Required	Laboratory's Graded Result		Analyte determined by state-certified lab
		Acceptable	Not Acceptable (Corrective Action Required)	
Minerals Alkalinity as CaCO ₃ Chloride Conductivity at 25°C Fluoride Potassium Sodium Sulfate Total Dissolved Solids at 180°C Total Solids at 105°C				
Hardness Total Suspended Solids Calcium Magnesium Calcium Hardness as CaCO ₃ Total Hardness as CaCO ₃ SM2340B	☐	☒	☐	☐
pH pH				
Settleable Solids Settleable Solids				
Solids Concentrate Total Suspended Solids Total Dissolved Solids at 180°C Total Solids at 105°C				
Solids Total Suspended Solids Total Dissolved Solids at 180°C Total Solids at 105°C				
Simple Nutrients Ammonia as N Nitrate + Nitrite as N Nitrate as N ortho-Phosphate as P				
Complex Nutrients Total Kjeldahl Nitrogen Total phosphorus as P				
Nitrite Nitrite as N				
Demand BOD CBOD COD TOC				

Permittee Name: Cadillac WWTP

Permit Number: MI0020257

EPA Lab Code: MI00041

Chemistry/Microbiology Analyte Checklist
DMRQA Study 37

Analyte Test / Method	Test Required	Laboratory's Graded Result		Analyte determined by state-certified lab
		Acceptable	Not Acceptable (Corrective Action Required)	
Oil & Grease Concentrate				
n-Hexane Extractable Material(O&G) (Grav)				
Oil & Grease				
n-Hexane Extractable Material(O&G) (Grav)				
n-Hexane Extractable Material(O&G) (IR)				
Trace Metals				
Aluminum				
Antimony				
EPA 200.7 & 200.8	☐	☒	☐	☐
Arsenic				
EPA 200.7 & 200.8	☐	☒	☐	☐
Barium				
EPA 200.7 & 200.8	☐	☒	☐	☐
Beryllium				
EPA 200.7 & 200.8	☐	☒	☐	☐
Boron				
Cadmium				
EPA 200.7 & 200.8	☐	☒	☐	☐
Chromium				
EPA 200.7 & 200.8	☐	☒	☐	☐
Cobalt				
Copper				
Iron				
Lead				
Manganese				
Molybdenum				
Nickel				
EPA 200.7 & 200.8	☐	☒	☐	☐
Selenium				
EPA 200.7 & 200.8	☐	☒	☐	☐
Silver				
EPA 200.7 & 200.8	☐	☒	☐	☐
Strontium				
Thallium				
EPA 200.7 & 200.8	☐	☒	☐	☐
Vanadium				
Zinc				
EPA 200.7 & 200.8	☐	☒	☐	☐
Low-Level Mercury				
Low Level Mercury				

Permittee Name: Cadillac WWTP

Permit Number: MI0020257

EPA Lab Code: MI00041

Chemistry/Microbiology Analyte Checklist

DMRQA Study 37

Analyte Test / Method	Test Required	Laboratory's Graded Result		Analyte determined by state-certified lab
		Acceptable	Not Acceptable (Corrective Action Required)	
EPA 1631E	☐	☒	☐	☐
Mercury				
Mercury				
Hexavalent Chromium				
Hexavalent Chromium				
Turbidity				
Turbidity				
Total Cyanide				
Cyanide, total SM4500CN C E	☐	☒	☐	☐
Amenable Cyanide				
Total Phenolics (4-AAP)				
Phenolics, total EPA 420.4	☐	☒	☐	☐
Total Residual Chlorine				
Total Residual Chlorine				
Low-Level Total Residual Chlorine				
Low Level Total Residual Chlorine				
WasteWatR™ Colliform MicrobE™				
Total Coliforms (MF) Fecal Coliforms (MF) E.coli (MF) Total Coliforms (MPN) Fecal Coliforms (MPN) E.coli (MPN)				
WasteWatR™ Colliform MicrobE™ - SM 9221				
Total Coliforms (MPN) Fecal Coliforms (MPN) E.coli (MPN)				

CINDY A. TOMASZEWSKI
Print Name
C. Tomaszewski, LAB. SUPERV.
Signature/Title

08-31-2017
Date

Use a separate checklist for EACH lab used

Permittee Name: Cadillac WWTP

Permit Number: MI0020257

EPA Lab Code: MI00082

Chemistry/Microbiology Analyte Checklist

DMRQA Study 37

Analyte Test / Method	Test Required	Laboratory's Graded Result		Analyte determined by state-certified lab
		Acceptable	Not Acceptable (Corrective Action Required)	
Minerals Alkalinity as CaCO ₃ Chloride Conductivity at 25°C Fluoride Potassium Sodium Sulfate Total Dissolved Solids at 180°C Total Solids at 105°C				
Hardness Total Suspended Solids SM 2540 D-1997 Calcium Magnesium Calcium Hardness as CaCO ₃ Total Hardness as CaCO ₃	☐	☒	☐	☐
pH pH SM 4500-H+ B-2000	☐	☒	☐	☐
Settleable Solids Settleable Solids				
Solids Concentrate Total Suspended Solids Total Dissolved Solids at 180°C Total Solids at 105°C				
Solids Total Suspended Solids Total Dissolved Solids at 180°C Total Solids at 105°C				
Simple Nutrients Ammonia as N SM 4500-NH ₃ D-1997 Nitrate + Nitrite as N Nitrate as N ortho-Phosphate as P	☐	☒	☐	☐
Complex Nutrients Total Kjeldahl Nitrogen Total phosphorus as P SM 4500-P E-1999	☐	☒	☐	☐
Nitrite Nitrite as N				

Permittee Name: Cadillac WWTP

Permit Number: MI0020257

EPA Lab Code: MI00082

Chemistry/Microbiology Analyte Checklist
DMRQA Study 37

Analyte Test / Method	Test Required	Laboratory's Graded Result		Analyte determined by state-certified lab
		Acceptable	Not Acceptable (Corrective Action Required)	
Demand BOD CBOD SM 5210 B-2001 COD TOC	☐	☒	☐	☐
Oil & Grease Concentrate n-Hexane Extractable Material(O&G) (Grav)				
Oil & Grease n-Hexane Extractable Material(O&G) (Grav) n-Hexane Extractable Material(O&G) (IR)				
Trace Metals Aluminum Antimony Arsenic Barium Beryllium Boron Cadmium Chromium Cobalt Copper Iron Lead Manganese Molybdenum Nickel Selenium Silver Strontium Thallium Vanadium Zinc				
Low-Level Mercury Low Level Mercury				
Mercury Mercury				
Hexavalent Chromium Hexavalent Chromium				
Turbidity Turbidity				

Permittee Name: Cadillac WWTP

Permit Number: MI0020257

EPA Lab Code: MI00082

Chemistry/Microbiology Analyte Checklist
DMRQA Study 37

Analyte Test / Method	Test Required	Laboratory's Graded Result		Analyte determined by state-certified lab
		Acceptable	Not Acceptable (Corrective Action Required)	
Total Cyanide				
Cyanide, total Amenable Cyanide				
Total Phenolics (4-AAP)				
Phenolics, total				
Total Residual Chlorine				
Total Residual Chlorine				
Low-Level Total Residual Chlorine				
Low Level Total Residual Chlorine				
WasteWatR™ Coliform MicrobE™				
Total Coliforms (MF)				
Fecal Coliforms (MF)				
SM 9222 D (m-FC)-2006				
E.coli (MF)				
Total Coliforms (MPN)				
Fecal Coliforms (MPN)				
E.coli (MPN)				
WasteWatR™ Coliform MicrobE™ - SM 9221				
Total Coliforms (MPN)				
Fecal Coliforms (MPN)				
E.coli (MPN)				

CINDY A. TOMASZEWSKI
Print Name
C. Tomaszewski, LAB. SUPERV.
Signature/Title

08-31-2017
Date

Use a separate checklist for EACH lab used

Permittee Name Cadillac WWTP State MI NPDES Permit No. 0020257 EPA Lab Code MI 01183

WET Analyte Checklist DMR-QA Study 37

Analyte Number	Organism / Conditions	Endpoint	Test Required	Laboratory's Graded Result		Analyte determined by state-certified lab*
				Acceptable	Not Acceptable (Corrective Action Required)	
Test Code 13 (refer to EPA Method 2000.0)						
754	Fathead minnow (<i>Pimephales promelas</i>) - MHSF 25°C	LC50	☒	☒	☐	☐
Test Code 14 (refer to EPA Method 2000.0)						
755	Fathead minnow (<i>Pimephales promelas</i>) - 20% DMW	LC50	☐	☐	☐	☐
Test Code 15 (refer to EPA Method 1000.0)						
756	Fathead minnow (<i>Pimephales promelas</i>) - MHSF	NOEC SURVIVAL	☒	☒	☐	☐
808	Fathead minnow (<i>Pimephales promelas</i>) - MHSF	IC25** (ON) GROWTH	☒	☒	☐	☐
810	Fathead minnow (<i>Pimephales promelas</i>) - MHSF	NOEC (ON) GROWTH	☒	☒	☐	☐
Test Code 16 (refer to EPA Method 1000.0)						
759	Fathead minnow (<i>Pimephales promelas</i>) - 20% DMW	NOEC SURVIVAL	☐	☐	☐	☐
812	Fathead minnow (<i>Pimephales promelas</i>) - 20% DMW	IC25** (ON) GROWTH	☐	☐	☐	☐
814	Fathead minnow (<i>Pimephales promelas</i>) - 20% DMW	NOEC (ON) GROWTH	☐	☐	☐	☐
Test Code 19 (refer to EPA Method 2002.0)						
764	<i>Ceriodaphnia dubia</i> - MHSF 25°C	LC50	☒	☒	☐	☐
Test Code 20 (refer to EPA Method 2002.0)						
765	<i>Ceriodaphnia dubia</i> - 20% DMW 25°C	LC50	☐	☐	☐	☐
Test Code 21 (refer to EPA Method 1002.0)						
766	<i>Ceriodaphnia dubia</i> - MHSF	NOEC SURVIVAL	☒	☒	☐	☐
767	<i>Ceriodaphnia dubia</i> - MHSF	IC25** REPRODUCTION	☒	☒	☐	☐
768	<i>Ceriodaphnia dubia</i> - MHSF	NOEC REPRODUCTION	☒	☒	☐	☐
Test Code 22 (refer to EPA Method 1002.0)						
769	<i>Ceriodaphnia dubia</i> - 20% DMW	NOEC SURVIVAL	☐	☐	☐	☐
770	<i>Ceriodaphnia dubia</i> - 20% DMW	IC25** REPRODUCTION	☐	☐	☐	☐
771	<i>Ceriodaphnia dubia</i> - 20% DMW	NOEC REPRODUCTION	☐	☐	☐	☐
Test Code 32 (refer to EPA Method 2021.0)						
788	<i>Daphnia magna</i> - MHSF 25°C	LC50	☐	☐	☐	☐
Test Code 38 (refer to EPA Method 2021.0)						
794	<i>Daphnia pulex</i> - MHSF 25°C	LC50	☐	☐	☐	☐
Test Code 42 (refer to EPA Method 2007.0)						
798	Mysid (<i>Americamysis bahia</i> , <i>Mysidopsis bahia</i>) 25°C	LC50	☐	☐	☐	☐
Test Code 43 (refer to EPA Method 1007.0)						
799	Mysid (<i>Americamysis bahia</i> , <i>Mysidopsis bahia</i>)	NOEC SURVIVAL	☐	☐	☐	☐
816	Mysid (<i>Americamysis bahia</i> , <i>Mysidopsis bahia</i>)	IC25** (ON) GROWTH	☐	☐	☐	☐
818	Mysid (<i>Americamysis bahia</i> , <i>Mysidopsis bahia</i>)	NOEC (ON) GROWTH	☐	☐	☐	☐
Test Code 44 (refer to EPA Method 2006.0)						
803	Inland silverside (<i>Menidia berylina</i>) 25°C	LC50	☐	☐	☐	☐
Test Code 45 (refer to EPA Method 1006.0)						
824	Inland silverside (<i>Menidia berylina</i>)	NOEC SURVIVAL	☐	☐	☐	☐
825	Inland silverside (<i>Menidia berylina</i>)	IC25** (ON) GROWTH	☐	☐	☐	☐
826	Inland silverside (<i>Menidia berylina</i>)	NOEC (ON) GROWTH	☐	☐	☐	☐
Test Code 46 (refer to EPA Method 2004.0)						
804	Sheepshead minnow (<i>Cyprinodon variegatus</i>) 25°C	LC50	☐	☐	☐	☐
Test Code 47 (refer to EPA Method 1004.0)						
805	Sheepshead minnow (<i>Cyprinodon variegatus</i>)	NOEC SURVIVAL	☐	☐	☐	☐
820	Sheepshead minnow (<i>Cyprinodon variegatus</i>)	IC25** (ON) GROWTH	☐	☐	☐	☐
822	Sheepshead minnow (<i>Cyprinodon variegatus</i>)	NOEC (ON) GROWTH	☐	☐	☐	☐

Name CINDY A. TOMASZEWSKI Signature CATomaszewski Date 08

*See Footnote on page 2.

**Preferred endpoint for DMR-QA performance test reporting.



A Waters Company

Study: DMR-QA 37
NPDES Permit #: MI0020257
Laboratory Name: Cadillac WWTP

Inorganic Results





DMR-QA 37 Final Report

A Waters Company

NPDES Permit #: MI0020257
Permit Holder: Cindy A Tomaszewski
Lab Supervisor
Cadillac WWTP
1121 Plett Rd.
Cadillac, MI 49601
(231) 775-2368

ERA Customer Number: C454401
Report Issued: 08/04/17
Study Dates: 03/24/17 - 07/07/17

TNI Analyte Code	Analyte	Units	Performance Evaluation	Reported Value	Assigned Value	Acceptance Limits	Method Description	Study Mean	Study Standard Deviation	USEPA Lab Code	Study
DMRQA Hardness (cat# 580, lot# Q037-507)											
1755	Total Hardness as CaCO ₃	mg/L	Acceptable	211.4	227	193 - 261	SM 2340B	224	9.49	MI00041	DMRQA37
DMRQA Hardness (cat# 580, lot# Q037-507)											
1960	Total Suspended Solids	mg/L	Acceptable	80.6	85.0	69.7 - 94.4	SM 2540 D-1997	81.3	3.20	MI00082	DMRQA37
DMRQA pH (cat# 577, lot# Q037-977)											
1900	pH	S.U.	Acceptable	5.54	5.54	5.34 - 5.74	SM 4500-H ₊ B-2000	5.53	0.0625	MI00082	DMRQA37
DMRQA Simple Nutrients (cat# 584, lot# Q037-505)											
1515	Ammonia as N	mg/L	Acceptable	16.9	17.4	14.0 - 20.6	SM 4500-NH ₃ D-1997	17.4	1.24	MI00082	DMRQA37
DMRQA Complex Nutrients (cat# 579, lot# Q037-525)											
1910	Total phosphorus as P	mg/L	Acceptable	5.42	5.79	4.80 - 6.71	SM 4500-P E-1999	5.87	0.332	MI00082	DMRQA37
DMRQA Demand (cat# 578, lot# Q037-516)											
1555	CBOD	mg/L	Acceptable	112	84.3	38.9 - 130	SM 5210 B-2001 2001	88.0	18.2	MI00082	DMRQA37



All analytes are included in ERA's A2LA accreditation. Lab Code: 1539-01
16341 Table Mountain Pkwy • Golden, CO 80403 • 800.372.0122 • 303.431.8454 • fax 303.421.0159 • www.eraqc.com





DMR-QA 37 Final Report

A Waters Company

NPDES Permit #: MI0020257
Permit Holder: Cindy A Tomaszewski
Lab Supervisor
Cadillac WWTP
1121 Platt Rd.
Cadillac, MI 49601
(231) 775-2368

ERA Customer Number: C454401
Report Issued: 08/04/17
Study Dates: 03/24/17 - 07/07/17

TNI Analyte Code	Analyte	Units	Performance Evaluation	Reported Value	Assigned Value	Acceptance Limits	Method Description	Study Mean	Study Standard Deviation	USEPA Lab Code	Study
DMRQA Trace Metals (cat# 586, lot# Q037-500)											
1005	Antimony	µg/L	Acceptable	83.6	96.0	61.6 - 126	EPA 200.7 & 200.8	95.9	5.21	MI00041	DMRQA37
1010	Arsenic	µg/L	Acceptable	247.4	249	201 - 295	EPA 200.7 & 200.8	251	14.3	MI00041	DMRQA37
1015	Barium	µg/L	Acceptable	2035	2120	1800 - 2440	EPA 200.7 & 200.8	2120	78.3	MI00041	DMRQA37
1020	Beryllium	µg/L	Acceptable	321	322	274 - 370	EPA 200.7 & 200.8	322	14.6	MI00041	DMRQA37
1030	Cadmium	µg/L	Acceptable	671.5	705	599 - 811	EPA 200.7 & 200.8	697	22.6	MI00041	DMRQA37
1040	Chromium	µg/L	Acceptable	516	519	441 - 597	EPA 200.7 & 200.8	521	22.4	MI00041	DMRQA37
1105	Nickel	µg/L	Acceptable	704.9	725	636 - 819	EPA 200.7 & 200.8	733	31.6	MI00041	DMRQA37
1140	Selenium	µg/L	Acceptable	460.1	475	404 - 546	EPA 200.7 & 200.8	474	24.7	MI00041	DMRQA37
1150	Silver	µg/L	Acceptable	501	513	436 - 590	EPA 200.7 & 200.8	513	23.9	MI00041	DMRQA37
1165	Thallium	µg/L	Acceptable	711.1	709	589 - 818	EPA 200.7 & 200.8	709	31.9	MI00041	DMRQA37
1190	Zinc	µg/L	Acceptable	1032	1100	935 - 1260	EPA 200.7 & 200.8	1100	49.1	MI00041	DMRQA37
DMRQA Low-Level Mercury (cat# 896, lot# Q037-931)											
1095	Low Level Mercury	ng/L	Acceptable	90.9	86.2	66.7 - 105	EPA 1631E	83.6	7.43	MI00041	DMRQA37
DMRQA Total Cyanide (cat# 588, lot# Q037-502)											
1645	Cyanide, total	mg/L	Acceptable	0.75	0.788	0.512 - 1.06	SM4500CN C E	0.768	0.0582	MI00041	DMRQA37
DMRQA Total Phenolics (4-AAP) (cat# 589, lot# Q037-515)											
1905	Phenolics, total	mg/L	Acceptable	0.782	0.563	0.277 - 0.849	EPA 420.4	0.624	0.111	MI00041	DMRQA37



All analytes are included in ERA's A2LA accreditation. Lab Code: 1539-01
16341 Table Mountain Pkwy • Golden, CO 80403 • 800.372.0122 • 303.431.8454 • fax 303.421.0159 • www.eraqc.com





A Waters Company

Study: DMR-QA 37
NPDES Permit #: MI0020257
Laboratory Name: Cadillac WWTP

Microbiology Results





A Waters Company

DMR-QA 37 Final Report

NPDES Permit #: MI0020257

Permit Holder: Cindy A Tomaszewski
Lab Supervisor
Cadillac WWTP
1121 Plett Rd.
Cadillac, MI 49601
(231) 775-2368

ERA Customer Number: C454401
Report Issued: 08/04/17
Study Dates: 03/24/17 - 07/07/17

TNI Analyte Code	Analyte	Units	Performance Evaluation	Reported Value	Assigned Value	Acceptance Limits	Method Description	Study Mean	Study Standard Deviation	USEPA Lab Code	Study
2530	Fecal Coliforms (MF)	CFU/100mL	Acceptable	450	1090	143 - 1530	SM 9222 D (m-FC)-2006	468	231	MI00082	DMRQA37

DMRQA WasteWater™ Coliform MicrobE™ (cat# 576, lot# Q037-083)

Per Section 6.4.3 a of the 2009 TNI Standard, "The assigned values for quantitative microbiology analytes shall be equal to the mean of the assigned value verification and/or homogeneity testing per Sections 7.1 and 7.2". The final acceptance limits are derived from the calculated study mean and study standard deviation from laboratory-reported results. Disagreement between the assigned values and study means/acceptance limits are due to the inherent variability of microbiology methods and differences in the methods used by ERA and participant laboratories. For quantitative microbiology analytes, the assigned value is not used in the evaluation of laboratories.



All analytes are included in ERA's A2LA accreditation. Lab Code: 1539-01
16341 Table Mountain Pkwy • Golden, CO 80403 • 800.372.0122 • 303.431.8454 • fax 303.421.0159 • www.eraqc.com

